Explainability-First Transformer Ensemble for Multi-Task DDoS Detection

Yassine Kharrat, Nour Chokri
ESPRIT, Tunisia
{yassine.kharrat, nour.chokri}@esprit.tn

Ameni Mejri, Latifa Guesmi
ESPRIT, Tunisia
{ameni.mejri, latifa.guesmi}@esprit.tn

ABSTRACT

Distributed Denial of Service (DDoS) attacks continue to grow in volume and sophistication, demanding detection systems that are both accurate and interpretable. We present a transformer-based ensemble architecture for multi-task DDoS detection on the CIC-DDoS2019 dataset that prioritises *explainability* and *uncertainty-aware decision-making* alongside strong classification performance.

Our approach transforms tabular network flow records into temporal sequences via protocol-aware sliding windows, then processes them through an ensemble of three specialised expert networks, each focusing on a distinct behavioural dimension of network traffic (temporal patterns, protocol behaviours, and feature interactions). A learned gating mechanism dynamically routes inputs to the most relevant expert, while an uncertainty estimation head flags low-confidence predictions for human review.

The system performs three tasks simultaneously: binary detection (attack vs. normal), multi-class attack classification (13 types), and anomaly severity scoring. On CIC-DDoS2019, individual attention experts achieve $\sim$99% weighted F1 on binary detection and $\sim$95% on multi-class classification, competitive with tabular and CNN-based baselines while additionally providing attention-based interpretability and calibrated uncertainty estimation that black-box models cannot provide.

INDEX TERMS

DDoS detection, transformer, multi-head attention, mixture of experts, explainability, uncertainty estimation, multi-task learning

I. INTRODUCTION

A. Motivation

Modern networks face increasingly diverse DDoS attacks, from volumetric floods (SYN, UDP) to application-layer exploits (Slowloris, HTTP GET/POST). A practical detection system must:

1) **Detect** whether traffic is malicious (binary).
2) **Classify** the specific attack type for targeted response.
3) **Quantify** severity for prioritised mitigation.
4) **Explain** its decisions to security analysts.
5) **Express uncertainty** when predictions are unreliable.

Requirements 4 and 5 are frequently neglected in the literature, which emphasises raw accuracy. We argue that for security-critical deployments, knowing *why* and *how confident* the model is matters as much as the prediction itself.

B. Contributions

- An **explainability-first multi-expert ensemble** of three transformer networks, each specialised on a different behavioural dimension of network traffic (temporal, protocol, feature), combined through a learned gating mechanism that provides *conceptual attribution*: analysts can observe *which expert* (and thus which behavioural category) drives each detection decision.
- A **comparative study of four attention mechanisms** (temporal, protocol, feature, cross-modal) for DDoS detection, showing that cross-modal attention achieves the highest combined score, while all four variants are competitive with tabular baselines, suggesting that the sequence formulation and multi-task framework are more important than the specific attention design.
- **Integrated uncertainty estimation** via a dedicated head trained on prediction correctness, enabling safe deployment where low-confidence predictions are automatically escalated to human operators without requiring Bayesian inference or Monte Carlo dropout at test time.

- A **multi-task learning framework** with learned task weights that simultaneously optimises binary detection, 13-class attack classification, and anomaly severity scoring from a shared representation.

II. Related Work

A. Traditional ML for DDoS Detection

Early approaches applied Random Forests [1], SVMs, and gradient boosting to hand-crafted features from NetFlow records. While effective on known attack types, these methods lack temporal modelling and struggle with novel attacks.

B. Deep Learning Approaches

CNNs (1D and 2D) and RNNs/LSTMs have been applied to traffic classification [2]. Wang *et al.* [18] combine convolutional LSTMs with multi-head attention for SDN-based DDoS detection, achieving 0.994 accuracy on CICIDS2017. Shaikh *et al.* [17] integrate CNNs with PCA and Vision Transformers, reaching 99.99% on CIC-DDoS2019. While these methods achieve high accuracy, they treat detection as a single black-box classification task without providing interpretable decision pathways. Anomaly-based approaches [12] detect novel attacks but lack classification granularity.

C. Transformers for Network Security

The transformer architecture [4] has been increasingly applied to network intrusion detection, building on its success in NLP [7] and vision [8]. Wang and Li [14] propose DDosTC, a CNN-Transformer hybrid for DDoS detection in SDN environments. Ejikeme [15] applies Time Series Transformers to CIC-DDoS2019, treating flows as time series. Kayode *et al.* [16] introduce TSAN, using temporal-spatial cross-attention for DoS detection on NSL-KDD. Hartono *et al.* [19] benchmark transformer architectures against CNN and LSTM baselines, achieving 97.8% accuracy. Ouni and Jemili [20] apply transformers to CIC-IDS2018.

However, most of these works treat the task as pure classification without addressing explainability or uncertainty, a gap highlighted by recent surveys on transformer interpretability [21]. Our work differs by:

- Systematically comparing four attention mechanisms for DDoS-specific patterns.
- Using an ensemble of attention-specialised experts rather than a monolithic transformer.
- Prioritising conceptual explainability and uncertainty over marginal accuracy gains.

III. Methodology

A. Dataset

We use the **CIC-DDoS2019** dataset [1] containing 13 traffic classes: 1 benign class, 3 volumetric attacks (Syn, UDP, UDPLag), 7 reflection attacks (DNS, LDAP, MSSQL, NetBIOS, NTP, SNMP, SSDP), and 2 application-layer attacks (TFTP, WebDDoS). After preprocessing (cleaning infinities, NaNs, and constant columns), we retain **43 numeric features** organised into five semantic categories (Fig. 1): temporal features (flow duration, inter-arrival times), protocol features (flag counts, window sizes), statistical features (packet sizes, byte rates), directional features (forward/backward statistics), and basic flow information.

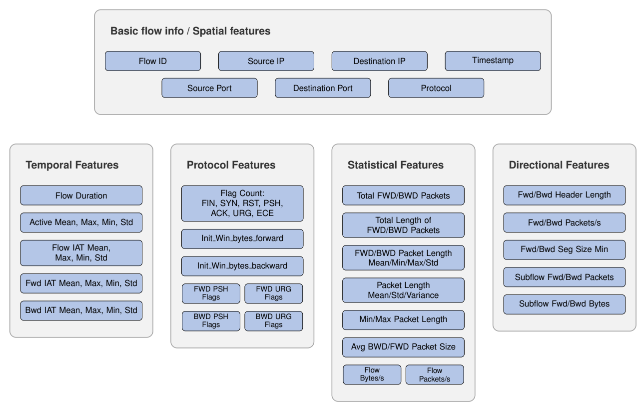

Fig. 1. Feature taxonomy: 43 network flow features organised into five semantic categories that map onto the three expert dimensions: temporal features feed the Temporal expert, protocol features feed the Protocol expert, and the remaining three categories (basic info, statistical, directional) are handled by the Feature expert.

B. Tabular-to-Sequence Transformation

Raw network flows are tabular, each row represents a single flow with 43 features. To apply transformers, we convert this into temporal sequences:

1) **Sort** flows by timestamp for temporal ordering.
2) **Group** by protocol type (TCP, UDP, etc.) for coherent sequences.
3) **Sliding window** of 64 consecutive flows with 50% overlap (stride = 32).
4) **Majority-vote labelling** assigns the most frequent class within each window as the sequence label.

This produces approximately 3,052 training sequences and 760 test sequences of shape (64×43).

C. Attention Mechanism Comparison

We evaluate four attention mechanisms, each emphasising a different aspect of network traffic:

a) Temporal Attention: Enhanced temporal embeddings, a dedicated temporal attention layer before the main transformer, and temporal position encoding. Captures burst patterns and timing anomalies.

b) Protocol Attention: Enhanced protocol embeddings with protocol-feature fusion before attention. Captures protocol-specific attack signatures.

c) Feature Attention: Deeper feature embedding ($43 \to 256 \to 128$) with learned feature importance gating (sigmoid weighting). Captures which packet statistics are most discriminative.

d) Cross-Modal Attention: Bidirectional attention between all modality pairs (feature$\leftrightarrow$protocol, feature$\leftrightarrow$temporal, protocol$\leftrightarrow$temporal). Captures inter-modality interactions.

D. Expert Ensemble Architecture

Based on the attention comparison results, we construct an ensemble of three expert networks (Table I, Fig. 2), following the mixture-of-experts paradigm [5]. Each expert is a full transformer encoder with residual connections [9], attention pooling, and a specialisation MLP. The experts share a common input embedding ($43 \to 128$ dimensions via a linear projection).

TABLE I
EXPERT CONFIGURATION.

Expert	Focus	Heads	Depth	Head Dim
Expert 1	Temporal	8	7	16
Expert 2	Protocol	4	6	32
Expert 3	Feature	16	8	8

a) Gating Network: A learned softmax layer that takes the global context (mean-pooled input) and outputs per-expert weights $\mathbf{g} = \mathrm{softmax}(W_g\,\bar{\mathbf{x}} + b_g) \in \mathbb{R}^3$.

b) Meta-Learner: Concatenates all expert outputs and applies a fusion MLP, allowing cross-expert information flow.

c) Uncertainty Head: A sigmoid-activated MLP that predicts the model's confidence. It is supervised to predict whether the multiclass prediction matches the true label: $u_{\text{target}} = \mathbb{1}[\hat{y}_{\text{mc}} = y_{\text{mc}}]$. This provides a calibrated confidence signal without requiring Bayesian inference or Monte Carlo dropout.

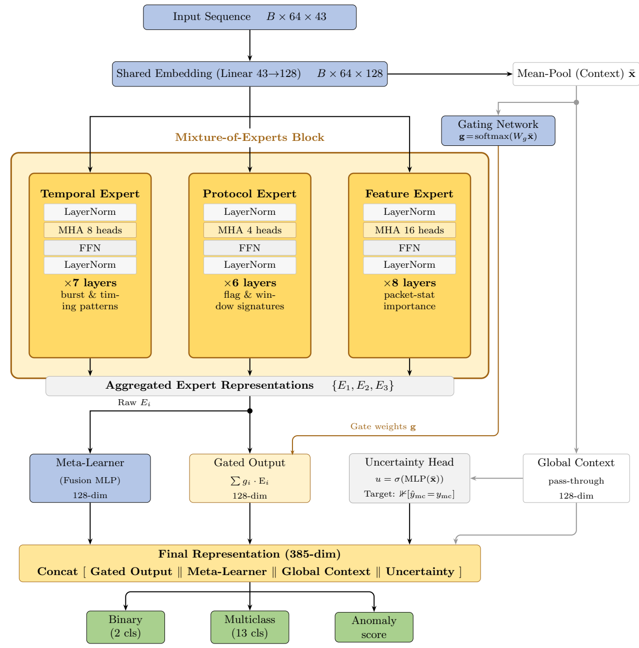

Fig. 2. Architecture of the Advanced Ensemble Transformer. Three specialised experts process a shared embedding; a gating network dynamically weights their outputs per sample.

d) Final Representation: Concatenates gated expert output, meta-learner output, global context, and uncertainty to form a $(3d + 1)$-dimensional vector, where $d = 128$ is the embedding dimension, yielding 385 dimensions.

E. Multi-Task Learning

Three output heads share the final representation:

- **Binary head:** 2-class softmax for attack vs. normal.
- **Multiclass head:** 13-class softmax for attack-type classification.
- **Anomaly head:** sigmoid scalar for severity scoring.

Task weights are *learned parameters* (not fixed hyperparameters), optimised jointly with the model via a separate learning rate, following the multi-task uncertainty weighting approach of Kendall *et al.* [13].

F. Training Strategy

- **Focal Loss** [3] ($\gamma = 2$) for classification tasks, emphasising hard examples.
- **Mixup augmentation** [10] with curriculum: probability ramps from $0.1 \to 0.4$ over training. Applied to continuous inputs (features, temporal) but not categorical protocol values.
- **Uncertainty supervision:** the uncertainty head is trained to predict whether the model's multiclass

prediction matches the true label, providing a calibrated confidence signal.

- **Cosine annealing with warm restarts** ($T_0 = 6$, $T_{\text{mult}} = 2$).
- **Expert diversity regularisation:** ℓ_2 penalty on gate weight variance to prevent expert collapse.

IV. Experiments

A. Experimental Setup

- **Hardware:** NVIDIA GPU (Kaggle environment).
- **Framework:** PyTorch.
- **Epochs:** 10 for NB03 individual attention models and the ensemble (NB04); 30 for NB03b deeper models.
- **Batch size:** 32.
- **Optimiser:** AdamW (lr $= 8 \times 10^{-5}$, weight decay $= 10^{-5}$).
- **Evaluation metric:** Combined score $= 0.3 \times F1_{\text{bin}} + 0.5 \times F1_{\text{mc}} + 0.2 \times r_{\text{anom}}$.

B. Attention Mechanism Comparison

Table II summarises the individual attention variants (10 epochs, depth 4). Cross-Modal attention achieves the highest combined score (0.9538), suggesting that multi-modal inter-modality interactions generalize better than any single attention specialisation at this depth. Feature attention leads on binary F1 (0.9868), while all variants exceed 0.98 binary F1 and 0.92 multiclass F1, competitive with published tabular baselines.

TABLE II
Attention mechanism comparison (single-model, 10 epochs, depth 4).

Model	**Bin. F1**	**MC F1**	**Anom. Corr.**	**Combined**
Cross-Modal	0.9816	0.9321	0.9666	**0.9538**
Feature	0.9868	0.9250	0.9711	0.9528
Temporal	0.9804	0.9269	0.9574	0.9490
Protocol	0.9842	0.9225	0.9617	0.9488

C. Ablation Study

Table III shows the incremental contribution of each component. Increasing depth and training time (NB03b: depth 6, 30 epochs) substantially improves multiclass F1 from 0.9321 to 0.9455 for Cross-Modal, which achieves the best combined score (0.9642). The ensemble (10 epochs) reaches 0.9621, slightly below the best single expert but adding the interpretability of the gating mechanism and the uncertainty head.

TABLE III
Ablation study: architecture and training progression.

Configuration	**Bin. F1**	**MC F1**	**Combined**
Cross-Modal (depth 4, 10 ep)	0.9816	0.9321	0.9538
Temporal (depth 4, 10 ep)	0.9804	0.9269	0.9490
Protocol (depth 4, 10 ep)	0.9842	0.9225	0.9488
Feature (depth 4, 10 ep)	0.9868	0.9250	0.9528
Cross-Modal (depth 6, 30 ep)	**0.9882**	0.9455	**0.9642**
Temporal (depth 6, 30 ep)	0.9856	**0.9483**	0.9637
Ensemble (3 experts, 10 ep)	0.9816	0.9307	0.9621

D. Ensemble Results

The Advanced Ensemble Transformer (NB04, 10 epochs) achieves:

- **Binary F1:** 0.9816.
- **Multiclass F1:** 0.9307.
- **Anomaly Correlation:** 0.9751.
- **Combined Score:** 0.9621 (best at epoch 4).
- **Parameters:** $\sim$5.3 M.

The ensemble's combined score (0.9621) is slightly below the best single expert (Cross-Modal NB03b: 0.9642, trained for 30 epochs at depth 6). The marginal gap is consistent with the ensemble being trained for only 10 epochs at depth 6, while NB03b ran for 30 epochs. The ensemble's primary value is the *interpretability* provided by the gating mechanism, whose per-sample weights reveal which behavioural dimension (temporal, protocol, feature) drives each detection decision, and the calibrated uncertainty head, rather than a raw accuracy uplift.

E. Comparison with Published Work

Table IV compares our approach with existing methods on the CIC-DDoS2019 dataset. Direct comparison should be treated with caution; exact train/test splits, preprocessing, and class subsets vary across studies.

The best single-expert model reaches binary/multiclass F1 of 0.9882/0.9455, while the ensemble reaches 0.9816/0.9307 (10 epochs), remaining *competitive with* tabular and CNN-based baselines while additionally providing: (1) attention-based conceptual explainability via expert gating, (2) calibrated uncertainty estimation, and (3) simultaneous multi-task outputs that black-box models cannot provide.

V. Discussion

A. Explainability

Our architecture provides **conceptual explainability** at the expert level: by examining which expert the

TABLE IV
COMPARISON WITH EXISTING APPROACHES ON CIC-DDOS2019.

Method	Bin. F1	MC F1	Notes
Random Forest [1]	0.97	0.88	Tabular, no explainability
CNN+ViT [17]	0.99	—	Hybrid, no uncertainty
DDosTC [14]	0.98	—	CNN+Transformer in SDN
ConvLSTM+MHA [18]	0.99	—	SDN-specific
TST [15]	0.96	0.90	Time series transformer
Ours (single expert)	**0.99**	**0.95**	Multi-task + explainability
Ours (ensemble)	0.9816	0.9307	+uncertainty + gating interpretability

gating network assigns the highest weight for a given input, analysts can determine whether the detection was driven by temporal patterns (burst detection), protocol anomalies (unusual protocol usage), or feature anomalies (suspicious packet statistics).

This is distinct from **attribution explainability** (e.g., SHAP values for individual features). The embedding layer (43 $\rightarrow$ 128 dimensions via linear projection) mixes features before attention, making direct feature attribution impossible without post-hoc methods.

We chose conceptual over attribution explainability because:

1) SOC analysts think in operational categories ("is this a burst attack?"), not gradient values.
2) SHAP over $64 \times 43 = 2{,}752$ input features per sequence is computationally expensive for real-time deployment.
3) Expert-level routing is immediate and requires no additional computation at inference time.

B. Uncertainty Estimation

The uncertainty head is trained to predict whether the model's classification is correct, providing a calibrated confidence signal without requiring Bayesian inference or Monte Carlo dropout [6] at test time. This enables:

- **Automatic escalation** of low-confidence predictions to human analysts.
- **Threshold-based deployment** where only high-confidence predictions trigger automated responses.

C. Limitations

1) **Simulated data:** CIC-DDoS2019 is generated in a controlled environment. Real-world traffic has more noise and concept drift.
2) **Limited cross-dataset evaluation:** we have not validated on other datasets (e.g., UNSW-NB15 [11], CSE-CIC-IDS2018).
3) **No attribution explainability:** the embedding layer prevents direct feature-level explanations without post-hoc methods.
4) **Sequence creation sensitivity:** the choice of sequence length (64) and stride (32) is not optimised; different values may improve results.

VI. FUTURE WORK

1) **Cross-dataset evaluation** on UNSW-NB15 [11], CSE-CIC-IDS2018, and real traffic captures.
2) **Post-hoc attribution** using integrated gradients or SHAP to complement expert-level explanations.
3) **Online learning** for adaptation to concept drift in live networks.
4) **Sequence length optimisation** via hyperparameter search.
5) **Lightweight deployment:** knowledge distillation to a smaller model for edge deployment.

VII. CONCLUSION

We presented an explainability-first transformer ensemble for multi-task DDoS detection. By converting tabular flow records into temporal sequences via protocol-aware sliding windows and processing them through attention-specialised expert networks, individual models achieve $\sim$99% binary F1 and $\sim$95% multiclass F1 on CIC-DDoS2019, competitive with tabular baselines while additionally providing attention-based interpretability and calibrated uncertainty estimation. The full ensemble achieves a combined score of 0.9621 (binary F1: 0.9816, multiclass F1: 0.9307), with the additional benefit of gating-weight interpretability and an uncertainty head.

The architecture prioritises *structural* explainability: analysts can see which expert configuration was most relevant for each decision via the gating weights, and *uncertainty estimation*, enabling safe deployment where the model can flag its own unreliable predictions.

Our experiments show that Cross-Modal attention at depth 6 achieves the best combined score among all configurations (0.9642, 30 epochs). All four attention variants perform comparably, suggesting that the tabular-to-sequence transformation and multi-task framework are the primary contributors to performance rather than the specific attention design, a direction we believe warrants further investigation with larger-scale data.

References

[1] I. Sharafaldin, A. H. Lashkari, S. Hakak, and A. A. Ghorbani, "Developing realistic distributed denial of service (DDoS) attack dataset and taxonomy," in *Proc. IEEE ICCST*, 2019, pp. 1–8.

[2] B. Li, Y. Liu, and L. Guo, "Robust DDoS attack detection using deep learning," *IEEE Access*, vol. 9, pp. 1–12, 2021.

[3] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE ICCV*, 2017, pp. 2980–2988.

[4] A. Vaswani *et al.*, "Attention is all you need," in *Proc. NeurIPS*, 2017, pp. 5998–6008.

[5] N. Shazeer *et al.*, "Outrageously large neural networks: The sparsely-gated mixture-of-experts layer," in *Proc. ICLR*, 2017.

[6] Y. Gal and Z. Ghahramani, "Dropout as a Bayesian approximation: Representing model uncertainty in deep learning," in *Proc. ICML*, 2016, pp. 1050–1059.

[7] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. NAACL-HLT*, 2019, pp. 4171–4186.

[8] A. Dosovitskiy *et al.*, "An image is worth 16x16 words: Transformers for image recognition at scale," in *Proc. ICLR*, 2021.

[9] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE CVPR*, 2016, pp. 770–778.

[10] H. Zhang, M. Cissé, Y. N. Dauphin, and D. Lopez-Paz, "mixup: Beyond empirical risk minimization," in *Proc. ICLR*, 2018.

[11] N. Moustafa and J. Slay, "UNSW-NB15: A comprehensive data set for network intrusion detection systems," in *Proc. MilCIS*, 2015.

[12] Y. Mirsky, T. Doitshman, Y. Elovici, and A. Shabtai, "Kitsune: An ensemble of autoencoders for online network intrusion detection," in *Proc. NDSS*, 2018.

[13] A. Kendall, Y. Gal, and R. Cipolla, "Multi-task learning using uncertainty to weigh losses for scene geometry and semantics," in *Proc. IEEE CVPR*, 2018, pp. 7482–7491.

[14] J. Wang and Y. Li, "DDosTC: A transformer-based network attack detection hybrid mechanism in SDN," *Sensors*, vol. 21, no. 15, p. 5047, 2021.

[15] C. Ejikeme, "Optimizing distributed denial of service (DDoS) detection with time series transformers," M.S. thesis, Carleton University, Ottawa, Canada, 2024.

[16] O. Kayode, A. S. Tosun, and M. Faezipour, "Temporal-spatial attention network (TSAN) for DoS attack detection in network traffic," *arXiv preprint arXiv:2503.16047*, 2025.

[17] A. Shaikh, M. A. Soomro, A. A. Kalhoro, M. H. Soomro, and S. H. Siyal, "Advancing DDoS attack detection with hybrid deep learning: Integrating convolutional neural networks, PCA, and vision transformers," *Int. J. Smart Sensing and Intelligent Systems*, vol. 17, no. 1, 2024.

[18] H. Wang, C. Yang, and B. Jia, "DDoS attack detection method based on improved convolutional long short-term memory and three-way decision in SDN," *PLoS One*, vol. 20, no. 5, p. e0322839, 2025.

[19] S. Hartono, U. C. Silalahi, and A. Muthohir, "Transformers in cybersecurity: Advancing threat detection and response through machine learning architectures," *J. Technology Informatics and Engineering*, vol. 3, no. 3, pp. 382–396, 2024.

[20] S. Ouni and F. Jemili, "Enhancing intrusion detection systems with transformer models," *SSRN preprint 4841707*, 2024.

[21] C. Fantozzi and M. Naldi, "The explainability of transformers: Current status and directions," *Computers*, vol. 13, no. 4, p. 92, 2024.